Mumbai, India | +91 77188 08655 | sandeep.pasup@gmail.com | https://www.linkedin.com/in/sandeep-pasupuleti

Sandeep Pasupuleti

Business Systems Analyst | GenAI, Healthcare IT, Airline Systems, Data Analytics, and Agile Delivery

Business Systems Analyst with 12+ years of experience across healthcare, airline, and semiconductor domains. Delivered GenAI solutions by defining requirements for conversational chatbot workflows and enabling data pipelines in collaboration with business and technical stakeholders. Experience includes data analysis using SQL and Python, with working knowledge of machine learning and data visualization for analytical decision-making.

Key Profile

- ✓ Led collaboration across product, engineering, and data science teams to design and implement AI-driven solutions, including GenAI chatbots, predictive models and process automation.
- ✓ Elicited requirements from business stakeholders and converted them into clear user stories and functional specifications that development teams could execute.
- ✓ Managed the entire project lifecycle for three major healthcare initiatives, including scope definition, regulatory compliance and production release.
- ✓ Integrated GDS and airline aggregation systems, enhancing booking workflows through Travel Fusion and Python Travel APIs.

Key Achievements & Impact

- ✓ Delivered a GenAI chatbot application for semiconductor manufacturing that enabled engineers to access process insights faster, resulting in a 40% improvement in information retrieval efficiency.
- ✓ Enhanced membership and enrollment systems for a major healthcare payer by reworking core business processes and improving data accuracy by 25%.
- ✓ Integrated machine learning-based predictive models into claims workfows that increased approval accuracy by 15% and reduced variability in outcomes.
- ✓ Automated workflows for airline booking and ticketing systems using Python, reducing manual errors and saving 100+ hours each month.

Core Competencies

- ☞ *Python Programming*
- ☞ *Data Analysis & SQL Query Writing*
- ☞ *Healthcare Domain Expertise*
- ☞ *Requirements Gathering*
- ☞ *Backlog Prioritization*
- ☞ *Machine Learning*
- ☞ *Data Visualization*
- ☞ *HIPAA Compliance*
- ☞ *User Story Creation*
- ☞ *Business Process Modeling*
- ☞ *GenAI Applications*
- ☞ *RAG Architecture*
- ☞ *Airline Distribution Systems*
- ☞ *Stakeholder Communication*
- ☞ *SDLC (Agile & Waterfall)*

Work Experience

Senior Business Systems Analyst | LAM Research (via DTC Infotech), Mumbai, India — May 2025 – Jan 2026

- ☞ Served as the Lead business analyst for Lambots, a RAG-based GenAI platform used by 500+ engineers globally to enable conversational chatbot use cases.
- ☞ Defined and documented functional requirements for data ingestion, embedding generation, multimodal indexing, and retrieval pipelines that guided GenAI solution delivery across 100+ structured and unstructured data sources.
- ☞ Established a benchmark testing approach with domain experts to validate LLM responses for accuracy and relevance, reducing hallucinations by 40%.
- ☞ Monitored adoption metrics, including user engagement, response quality, and satisfaction scores.
- ☞ Built executive dashboards that demonstrated 85% user satisfaction and informed product roadmap decisions.
- ☞ Improved model response quality by incorporating user feedback and testing results, leading to a significant reduction in the volume of irrelevant responses.

Lead Business Systems Analyst | Cognizant (via Tquanta Technologies), Mumbai, India — Nov 2024 – Apr 2025

- ☞ Drove Agile delivery for Medicaid-related FACETS system enhancements by creating Epics, refining the backlog, and coordinating sprint priorities with development, QA and business teams.
- ☞ Maintained Jira health through structured user story management, sprint tracking, and facilitation of daily stand-ups and sprint reviews.
- ☞ Analyzed and standardized Medicaid enrollment workflows within the FACETS system to maintain accurate member records for over 1 million beneficiaries.
- ☞ Conducted detailed design sessions, bridging gaps between business needs (Illinois Medicaid) and technical execution, resulting in 100% on-time epic closures and production deployment.

Senior Consultant | Qatar Airways (via Virtusa Consulting Services), Doha, Qatar — May 2023 – Jul 2024

- ☞ Partnered with Qatar Airways' Group Medical Division to assess gaps in the Hospital Management System (HMS) and implement enhancements that improved day-to-day operations and system usability.
- ☞ Led the creation of Business Requirements Document (BRD) and Functional Requirements Document (FRD), aligning system enhancements with business needs identified by clinical and admin users.
- ☞ Performed SQL-based data analysis on clinical datasets, with Python used for data validation and trend analysis to identify workflow inefficiencies.
- ☞ Ensured regulatory compliance with Health Level Seven (HL7), Digital Imaging and Communications in Medicine (DICOM), General Data Protection Regulation (GDPR) standards, maintaining 100% compliance across implemented changes.
- ☞ Automated manual processes at Qatar Airways Medical Centre, reducing data entry errors by 65% and improving overall data accuracy.

Senior Business Analyst | Conduent (via Collabera Technologies), Bangalore, India Sep 2022 – Mar 2023

- Served as the primary point of contact between business stakeholders and the development teams for Conduent's Drug Rebate Administration platform, translating complex pharmaceutical rebate rules into BRD and FRD documentation.
- Conducted advanced SQL analysis on drug rebate data and pricing data to support operational improvements.
- Translated pharmacy claim adjudication rules and data attributes into input used for machine learning-based predictive model.
- Led code review coordination and supported production rollouts, reducing deployment issues by 25%.
- Mentored three QA team members to identify and resolve critical defects, improving test case coverage by 20% prior to release.

Senior Business Analyst | UST Global, Bangalore, India Jun 2019 – Sep 2022
Client Engagements: Amadeus Software Labs, Legato Health Technologies

- Delivered airline customer integrations for Amadeus Python Flight portal through structured implementation planning and stakeholder coordination, achieving 98% client satisfaction.
- Managed end-to-end implementation of 20+ airline integrations with Amadeus solutions, coordinating project timelines and technical deliverables.
- Analyzed airline API requests and responses (XML/JSON) to improve integration efficiency through better data mapping, feature extraction and web scraping workflows.
- Applied Python-based automation to booking and ticketing processes, reducing manual intervention.
- Led collaboration with the R&D team on product delivery and feature validation, ensuring configurations aligned with customer requirements.
- Reduced data discrepancies by 20% by executing complex SQL queries for in-depth data analysis, delivering actionable insights.

Additional Experience

Business Analyst | Mphasis, Bangalore, India Jan 2018 – Apr 2019

Systems Analyst | UnitedHealth Group (via ITC Infotech), Hyderabad, India Nov 2016 – May 2017

Business Analyst | 3R Consulting LLC, New Jersey, USA Jan 2012 – Aug 2016
Client Engagements: BCBS Michigan, Horizon BCBS New Jersey

Education

Bachelors in Electronics and Communications Engineering 2008
Jawaharlal Nehru Technological University, Hyderabad, India

Certifications

Data Scientist: Machine Learning Certification | Codecademy 2024

- Comprehensive training in Python programming, data analysis, hypothesis testing, machine learning, feature engineering, model building, model evaluation, and deep learning with neural networks.

Technical Skills

- **Languages & Frameworks**: Python, SQL, TensorFlow, PyTorch, Scikit-Learn, Transformer Models
- **Data Extraction**: Web Scraping (BeautifulSoup, Scrapy)
- **Data Analysis & Visualization**: Pandas, NumPy, Power BI, Matplotlib, Plotly
- **Machine Learning & AI**: Deep Learning, NLP, LLMs, RAG, Embeddings & Vector-Based Retrieval, Model Evaluation & Validation
- **Statistical Methods**: Exploratory Data Analysis, Probability Distributions, Hypothesis Testing, A/B Testing
- **Collaboration Tools**: Jira, Confluence, Azure DevOps

Certification Projects

AI-Powered Financial Chatbot

- Built an NLP-driven chatbot using BERT for intent classification and spaCy for entity extraction. Implemented LSTM-based time series forecasting for financial metric predictions with integrated data analysis features including metric comparison, ratio calculations, and growth rate analysis.

Airline Customer Review Analysis and Predictive Modeling

- Developed a web scraping pipeline to collect and analyze airline review data, applying NLP techniques for topic modeling and sentiment analysis. Built a Machine Learning pipeline using Random Forest and XGBoost in a stacking ensemble, achieving a 63% recall in predicting customer purchases and a 76% weighted F1-score. Applied advanced data preprocessing techniques, including SMOTEENN, to handle imbalanced data and improve model performance.

Customer Churn Prediction Using Imbalanced Data Techniques

- Built a customer churn prediction model leveraging ensemble methods including Balanced Random Forest, XGBoost, and LightGBM, optimized via RandomizedSearchCV. Addressed class imbalance using SMOTETOMEK and performed feature selection with RFECV. Employed model stacking and calibration, achieving a weighted F1 score of 87% with a balanced trade-off between precision and recall.